



# Haptic Graphics Rendering for Blind and Low Vision Students

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## Abstract

We present a haptic mouse for tactile graphic rendering to assist with remote learning of blind and low vision (BLV) students. The haptic mouse provides kinesthetic feedback and consists of a 3 DOF (height, pitch, yaw) modified Stewart platform. We discuss the system's design, integration, and preliminary implementation. Finally, we propose the future integration of a voice coil actuator (VCA) for haptic texture rendering and a sensor for absolute position tracking.

## Motivation

- Blind and low vision (BLV) students use tactile graphics to understand non-textual information, such as graphs or pictures.
- Tactile graphics literacy is especially important for STEM subjects where diagrams and graphs are common.
- Current strategies often rely upon one-on-one guidance with a sighted teacher [1]. However, physical interaction is not possible i.e., remote learning.



Figure 1(a): A tactile graphic depicting North America.

## Related Work

- Haptic guidance using skin-stretch haptics and multimodal (i.e., audio and kinetic) feedback [2,3].
- Tiltable platforms, such as *NormalTouch*, are effective in conveying surface normals for high fidelity virtual object rendering [4].
- Augment graphics rendering by leveraging voice coil actuators (VCAs) to render virtual textures [5].

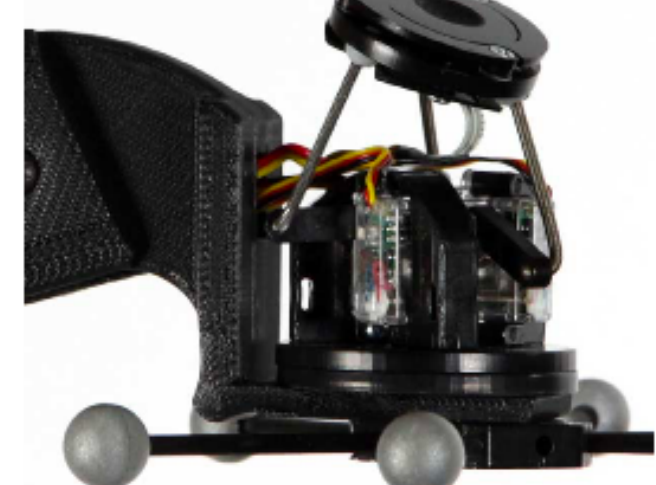


Figure 1(b): *NormalTouch* a tiltable and extrudable handheld haptic device.

## Objective

- Design a haptic mouse to support BLV students as they learn to explore 3D graphics.

## Design Considerations

We are motivated to ensure that tactile graphics training can be made widely available for BLV people. Our design considerations were guided by the principles of ability-based design [6] and are as follows:

- Ability-centered (i.e., leveraging haptic feedback)
- Open source
- Low cost, easily replicable and adaptable

## System Design

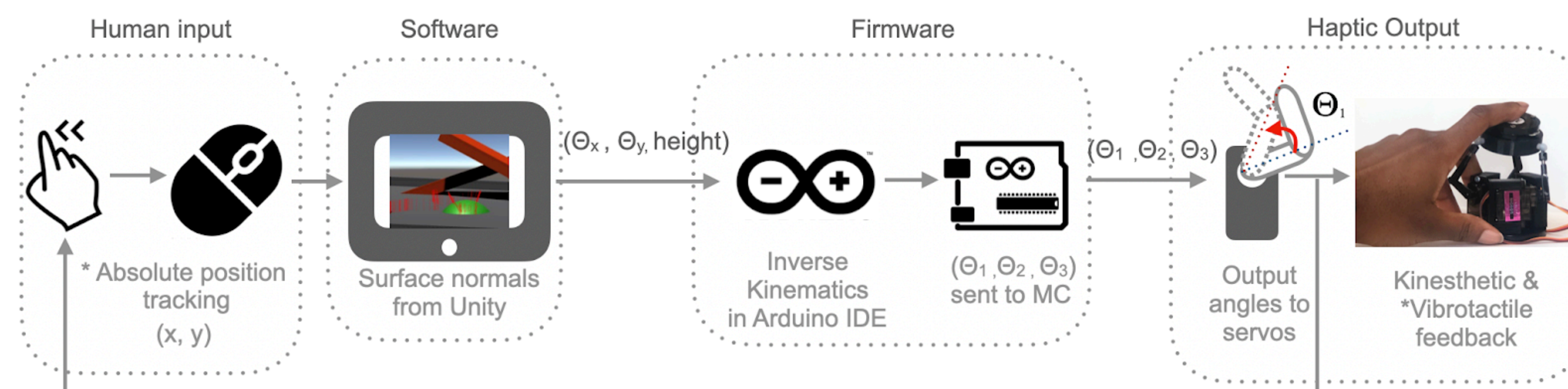


Figure 2(a): Haptic mouse system design  
Note: \* elements that have not been implemented in this prototype

- Project scope:** design, system integration, and functionality testing.
- Perception focus:** One hand. Single finger for simplified stimulus induced shape perception [7].
- Fig 2(a) summarizes our system flow.
- My primary focus was the **device kinematics** and **mechanical design**.

## I. Device Kinematics

**Objective:** Robotic control and workspace optimization to maximize the capabilities of our hardware and allow it to render a wide range of surfaces.

**Methods:**

**A. Deriving inverse kinematic equations for tilt platform control.**

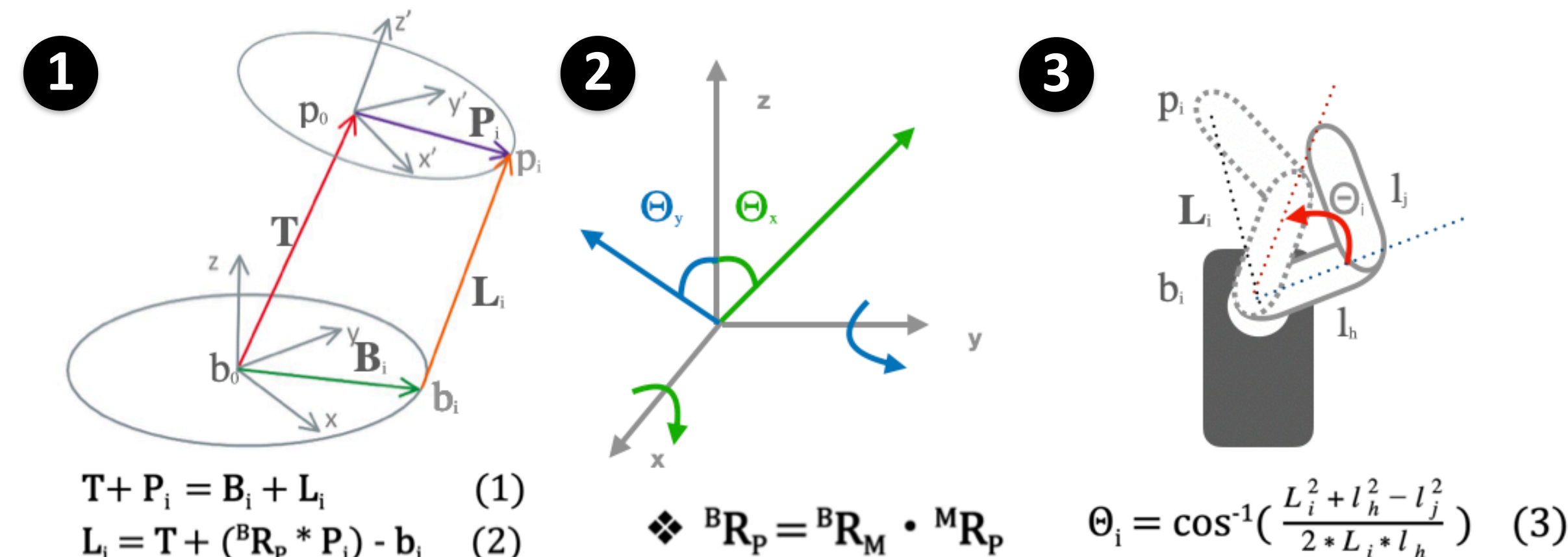


Figure 2(b): Device Kinematic:  
(1) Platform & Vector Definitions; (2) Rotation Matrix; (3) Servo Angles

**B. Writing an algorithm that determines the reachable workspace based on random combinations of height, and orientation angles.**

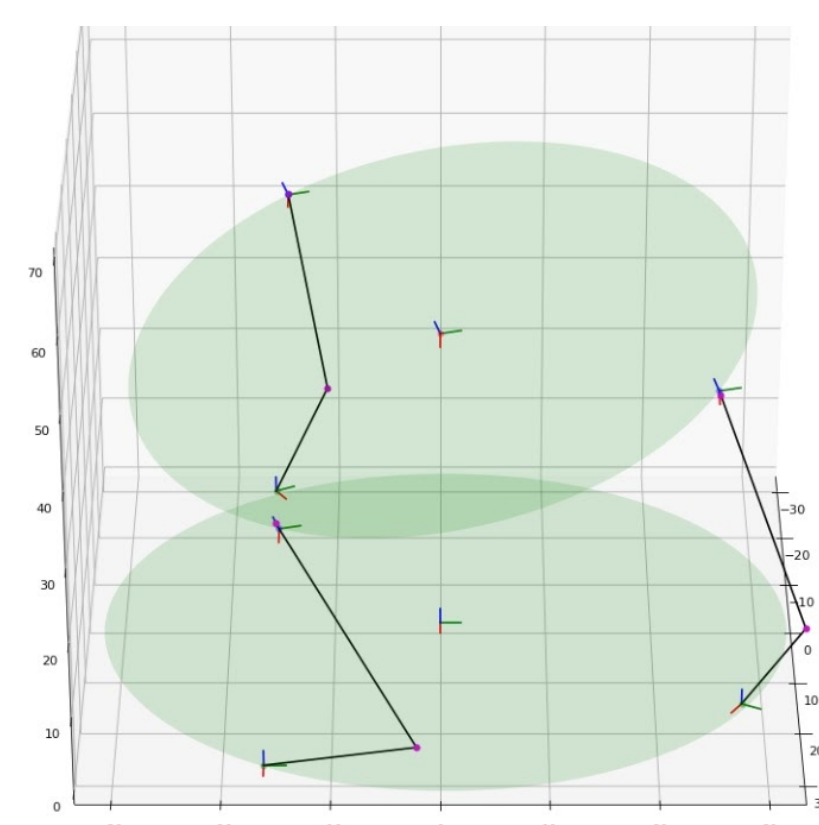


Figure 2(c): Device Simulation of Inverse Kinematics for  $\theta_x = 15$ ,  $\theta_y = 15$ , height = 40

**Results:**  
• 50 % drop in initial linkage length results in a workspace increase of 30%.

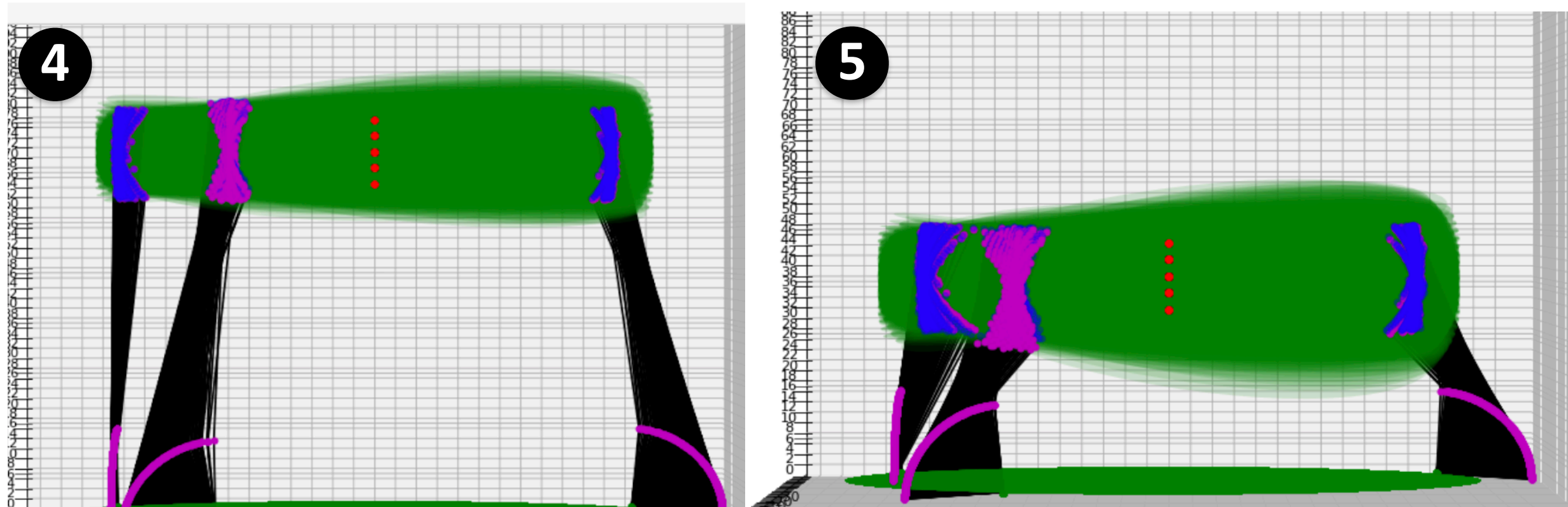


Figure 2(d): Device Simulation for Workspace Optimization for:  
(4) initial linkage length of 61.38mm; (5) final linkage length of 30.69mm

## II. Mechanical Design

- Components:** three externally-powered 6V MG90S 180-degree servo motors & Arduino Elegoo Uno R3 microcontroller.
- Fabrication:** laser cut acrylic & 3D printed in PLA.
- Firmware:** programmed in Arduino IDE.
- Software:** virtual environment developed in Unity.

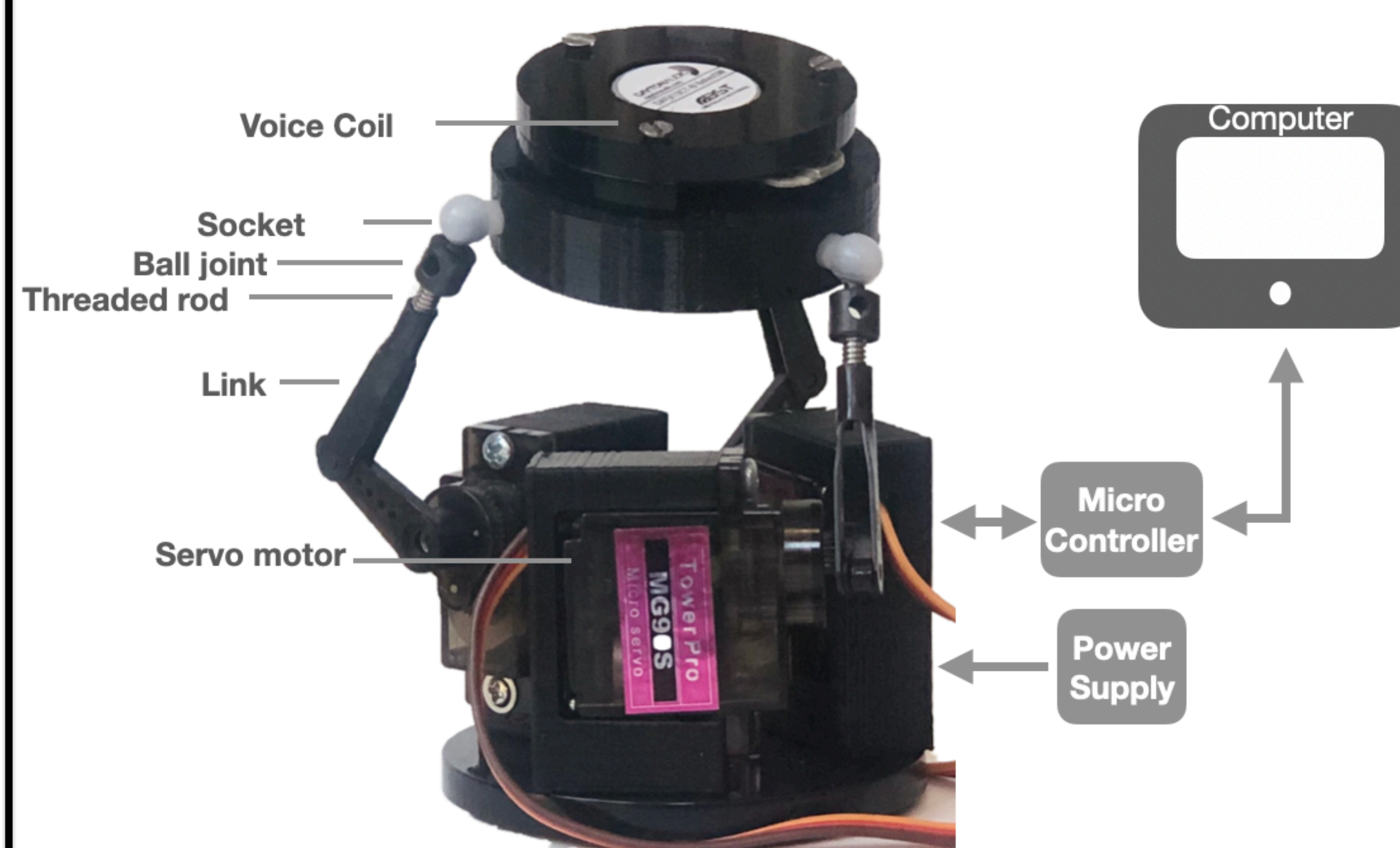


Figure 2(e): Physical Device showcasing hardware components and system integration.

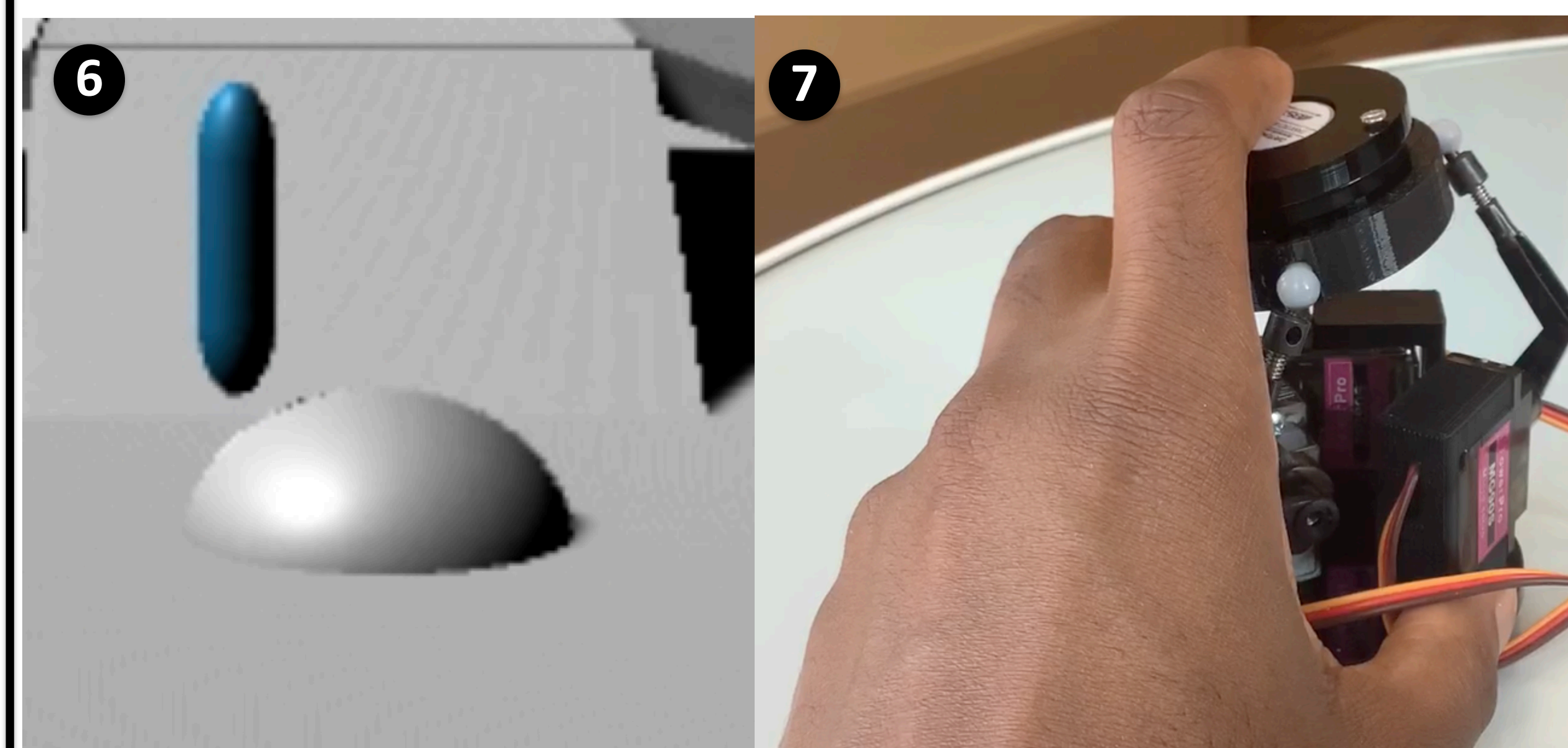


Figure 2(f): Haptic Rendering of Curvature  
(6) Moving over virtual environment; (7) Corresponding physical movement

## Limitations

- Limited to kinesthetic haptic feedback. The voice coil actuator was not integrated for this prototype.
- Feedback was limited to single hand. Bimanual exploration is common strategy for tactile graphics and will be explored in future prototype.
- Maximum rendered angle between 20 - 25 degrees.
- No 2D position input from sensor to enable our device to function similar to that of a mouse

## Future Work

- Incorporate voice coil actuator to augment haptic fidelity through virtual texture rendering.
- Incorporate absolute positioning for 2D position tracking and to reduce position mismatch between the virtual representations and physical outputs [8].
- Develop tactile graphics training applications and user test.

## Acknowledgements

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